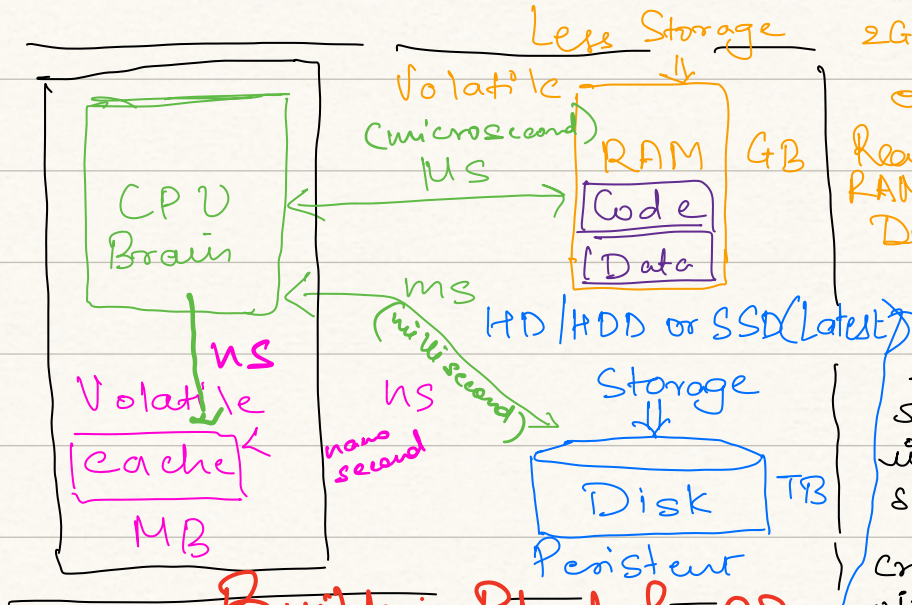


Computer Architecture

Random Access Memory
2GB/4GB/6GB...
32
expensive
Read/Write in RAM faster than Disk
 $\mu s = \frac{1}{10^6}$



Building Block for SD

How much can a disk store?

[1GB = 10^9] 100 GB or 1TB [1TB = 10^{12} Bytes] Read/Write in $ms = \frac{1}{10^3}$

1 Byte = 8 Bits [00100001]
Persistent storage

If we store anything in disk and say if our computer crashes all of it can be recovered through disk.

1) Smaller than disk & RAM data in MB

2) Used to speed up Read/Write operation from RAM in $ns = \frac{1}{10^9}$

3) Cache used as substitution of RAM

4) Use of cache if there is a portion/data from RAM that is being read frequently it will be

stored in Cache so as to optimize read operation [So we first go to the cache and see if data is there if not there then we do the expensive operation of reading from RAM]

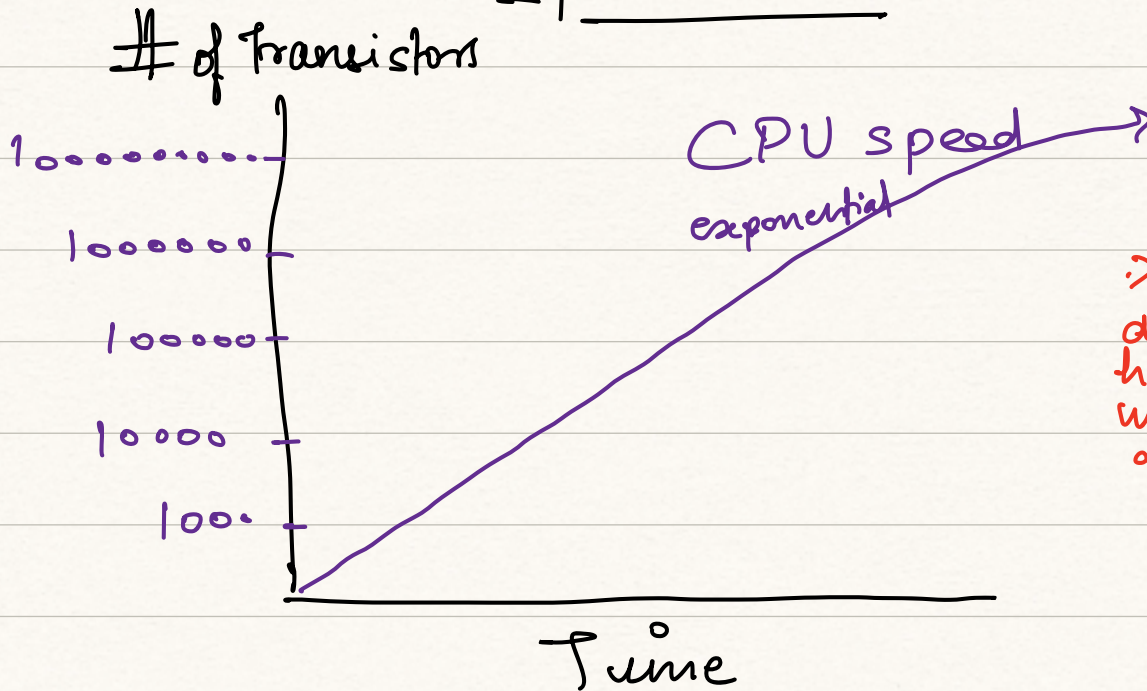
Conclusion :- So if we want to solve complex problems we need to use the above Sys or Scale

Data Storage	Units
Bit	1 or 0
Byte	8 bits
Kilobyte	1,000 bytes
Megabyte	1,000 kilobytes
Gigabyte	1,000 megabytes
Terabyte	1,000 gigabytes
Petabyte	1,000 terabytes
Exabyte	1,000 petabytes
Zettabyte	1,000 exabytes
Yottabyte	1,000 zettabytes

$\rightarrow 10^3$ bytes
 $\rightarrow 10^6$ bytes
 $\rightarrow 10^9$ bytes
 $\rightarrow 10^{12}$ bytes

it if needed.

Moore's Law



CPU speed
slowing
down for
1 computer

∴ Indiv comp
do have limitation
hence need
multiple CPU or
distributed
system